Tribhuvan Univesity
Institute of Science and Technology
2071

Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology Stat. 108
(Statistics I)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
All notations have the usual meanings.

Group A

Attempt any Two: (2×10=20)

1. Describe in detail simple random sampling method for drawing a random sample of size n from a population size N.

   Show that $\left(V_{srswr} - V_{srswor}\right) = \frac{n-1}{Nn} S^2$ where V_{srswr} denote the variance of unbiased estimator of the population mean under simple random sampling with replacement and V_{srswor} denote the variance of unbiased estimator of the population mean under simple random sampling without replacement. Also give your comments based on this result.

2. Explain the need of Kruskal Wallis one way ANOVA test.
   A researcher wants to find out if there is any difference in the level of morale among various Nepal Government computer engineer. Eighteen computer engineers were selected at random from three Ministries of Nepal Government and their morale level was established by using standard predetermined scale. The values so determined for each Ministry are given below.

Ministry	Finance	65	68	70	73	74	77	70
	Education	60	62	64	67	67	71	
	Foreign Affairs	74	63	66	69	67		

   Use the Kruskal Wallis one way ANOVA test at 5% level of significance to test the hypothesis that the morale level of the computer engineers of three ministries is the same.

3. A random sample of 25 nurses selected from a state registry of nurses yielded information on each nurse's score on the state board examination (Y) and his/her final score in school (X_1). Both scores relate to the nurse's area of affiliation. Additional information on the score made by each nurse on an aptitude test (X_2), taken at the time of entering nursing school, was made available to the researcher. To study the effect of final score (X_1) and aptitude test score (X_2) on the state board examination score (Y), the following estimated regression model based on data is given below.

   $\sum\left(Y_1 - \bar{Y}\right)^2 = 55300.000$ and $\sum\left(Y_1 - \hat{Y}_1\right)^2 = 38282.215$ and model is fitted as:

   $$\hat{Y} = -421.997 + 11.166\ X_1 - 0.630\ X_2$$

   (SE) (339.762) (3.655) (0.938)

   (i) Explain and interpret the estimated regression coefficients of the fitted model.
   (ii) Compute the value of R^2 and its adjusted value and also give your comments on it.
   (iii) Obtained the predicted value of Y when $X_1 = 98$ and $X_2 = 90$.
   (iv) Test the significance of slop regression coefficients at 5% level of significance.
   (v) Carry out the overall goodness of fit test of the model at 5% level of significance.

Group B

Answer any eight questions: (8×5=40)

4. Describe in detail multistage sampling method with illustrations.

5. A sample survey was conducted to estimate area under rice production in a Village Development Committee. The following data were obtained.

No. of stratum	Total no. of holdings (N_n)	No of sample unit (n_h)	Mean area under rive per holding ($(\bar{y}_h)$)	SD of area under rice per holding (s_h^2)
1	125	2	3	0.6
2	130	3	4	0.4
3	145	4	5	0.5
4	150	5	6	0.4
5	165	6	7	0.3

Find the sample estimate of population mean and population total, variance of the estimate of sample mean and estimate of the total variance, and also estimate the standard error of $\bar{y}_{st}$.

6. Write down the rationale and method of Wilcoxon matched pairs signed rank test.

7. A survey reported the level of mental health scores of the primary school girl students from hill and tarai belts in given below:

Mental health scores: hill:	29	50	43	22	42	47	42	32
Mental health scores : tarai:	35	26	27	51	27	38	23	25
	25	27	45					

Use the Median test to carry out the test statistics at 5% level of significance whether there is a difference of the two mental health scores of the girl students from hill and tarai or not.

8. Describe the method of Kolmogorov Smirnov test in case of one sample.

9. The following data comes from a random sample of seven industrial firms.

n	1	2	3	4	5	6	7
Output (Q)	100	120	140	150	165	190	200
Labour (L)	1.0	1.3	1.8	2.0	2.5	3.0	4.0
Capital (K)	2.0	2.2	2.3	1.5	2.8	3.0	3.4

Fit the Cobb-Douglas production function and interpret the results.

10. Discuss the problems of multicollinearity. What measures do you suggest to remove the effect of multicollinearity?

11. Explain the autoregressive model for first order and also discuss the problems involved in estimating such model.

12. Define partial correlation coefficient. Also, state the assumptions and properties of it. Given that the simple correlation coefficient $r_{12} = 0.53$, $r_{13} = 0.35$ and $r_{23} = 0.85$, compute $r_{12.3}$, $r_{23.1}$ and coefficient of partial determination.

13. Write short notes of **any two**:
    (a) Sign test
    (b) Growth model
    (c) Autocorrelation in the regression estimators